

Random Walk VBA Program Documentation

Objective

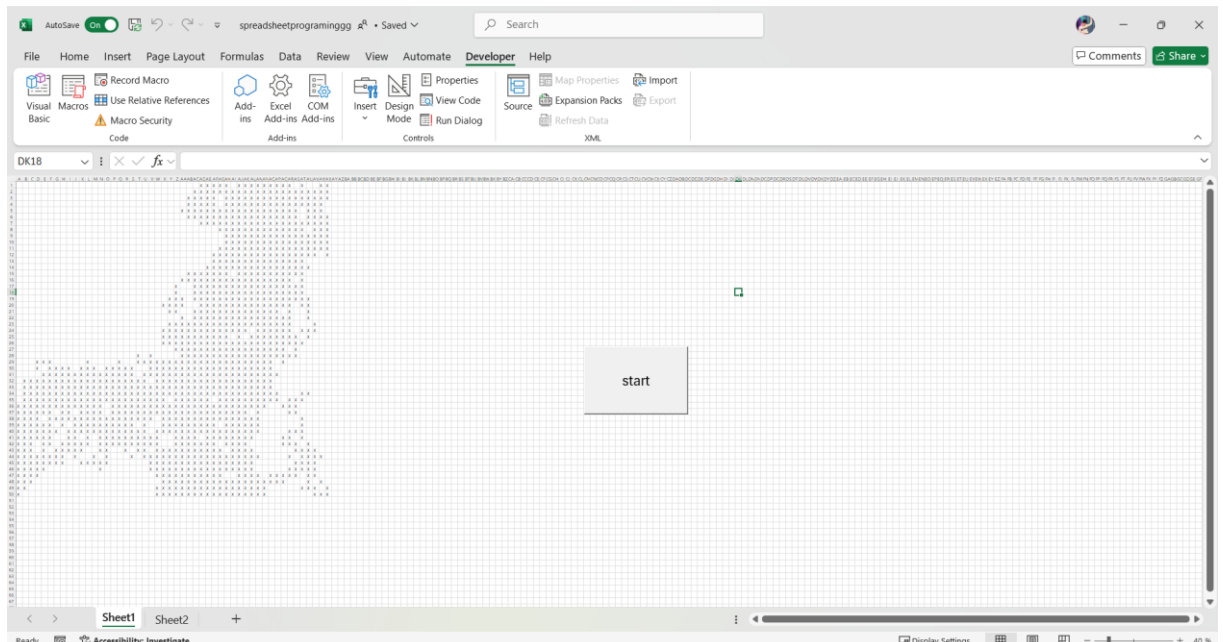
The **Random Walk** program simulates random path movement within a grid using Excel's VBA functionality. Users can customize the movement dimensions (1D, 2D, or 8D), grid size, and additional options such as revisiting and wrap-around functionality to control how the path behaves on the grid.

Features

- **Movement Type:** Users can choose between 1D, 2D, or 8D random walks:
 - **1D:** Movement occurs diagonally (up-right or down-right). The walk alternates between moving one step right and either up or down.
 - **2D:** Movement occurs in two directions (up, down, left, right).
 - **8D:** Movement occurs in eight possible directions: North (N), North-East (NE), East (E), South-East (SE), South (S), South-West (SW), West (W), and North-West (NW).
- **Step and Grid Size:** Users can define the number of steps and the grid dimensions. This determines the path length and size of the simulation area.
- **Revisit and Wrap-Around Options:** Users can choose:
 - **Allow Revisits:** Whether the walker is allowed to visit previously visited points.
 - **Wrap-Around:** Whether the walker should continue moving in the opposite direction if they reach the boundary of the grid.
- **Graphical Representation:** The random walk path is displayed visually within an Excel sheet, where each step is represented by a cell.
- **UserForm Interface:** The program utilizes a UserForm to allow for easy configuration of the simulation parameters. The UserForm contains input fields for the number of steps, grid size, movement type selection, and checkboxes for revisits and wrap-around functionality.

How to Use

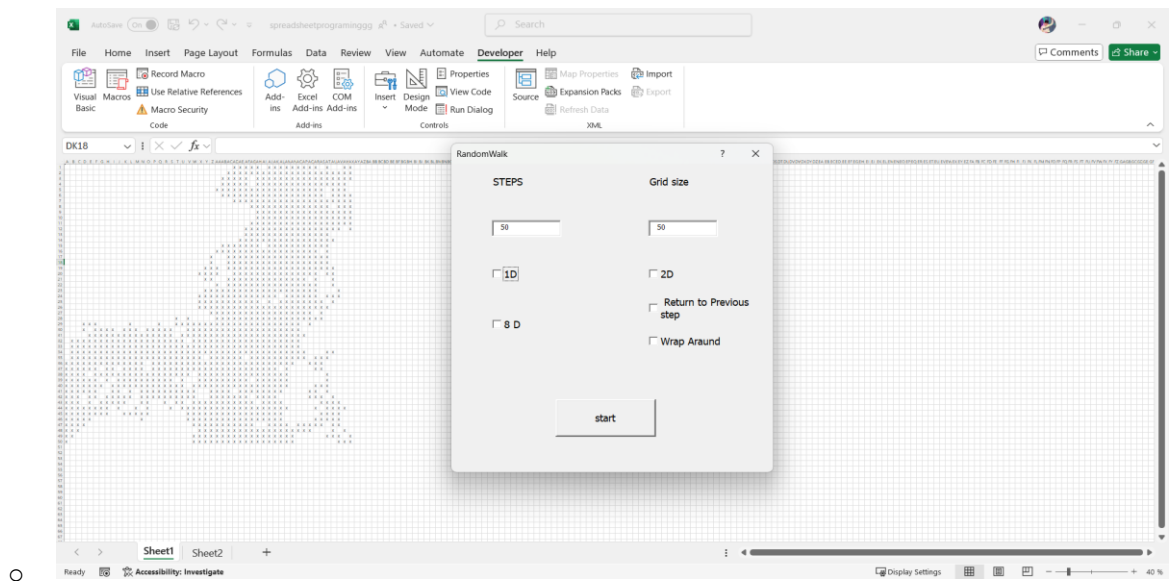
1. **Open the Excel File:** Ensure you have the Excel workbook containing the Random Walk program.



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3. Click the Excel Start Button:

- First, click the **Start** button located in the Excel sheet. This will open the **UserForm**, where you can configure your simulation parameters.



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4. Configure Simulation in the UserForm:

- Number of Steps:** Enter the total number of steps for the walker to take.
- Grid Size:** Define the grid's dimensions (width and height).
- Movement Type:** Select the movement type (1D, 2D, or 8D).
- Revisits:** Choose whether the walker is allowed to revisit previously visited cells.

- **Wrap-Around:** Select whether the walker should wrap around when reaching the grid boundaries.

The screenshot shows a window titled "RandomWalk" with a standard Windows-style title bar (minimize, maximize, close buttons). The window is divided into two main sections: "STEPS" on the left and "Grid size" on the right. In the "STEPS" section, there is a text input field containing "5000". Below it are three checkboxes: "1D" (unchecked), "2D" (unchecked), and "8 D" (checked). In the "Grid size" section, there is a text input field containing "100". Below it are three checkboxes: "Return to Previous step" (checked), "Wrap Araund" (checked), and "2D" (unchecked). At the bottom center of the window is a large button labeled "start".

5. **Click the UserForm Start Button:**

- Once you've entered all your parameters, click the **Start** button inside the **UserForm**. This will begin the random walk simulation using the entered configuration.

6. **Grid Setup:**

- The program will clear any existing data in the Excel grid and create a clean grid with the size you defined.

7. **Random Walk Path Generation:**

- The walker will begin the random walk, taking steps according to the movement type selected (1D, 2D, or 8D).

- Each step will be marked with an "X" in the grid, visually representing the path of the walker.

8. Wrap-Around (if enabled):

- If you chose the wrap-around option, the program will check whether the walker has reached the edge of the grid. If so, the walker will move to the opposite side of the grid, simulating a continuous loop.

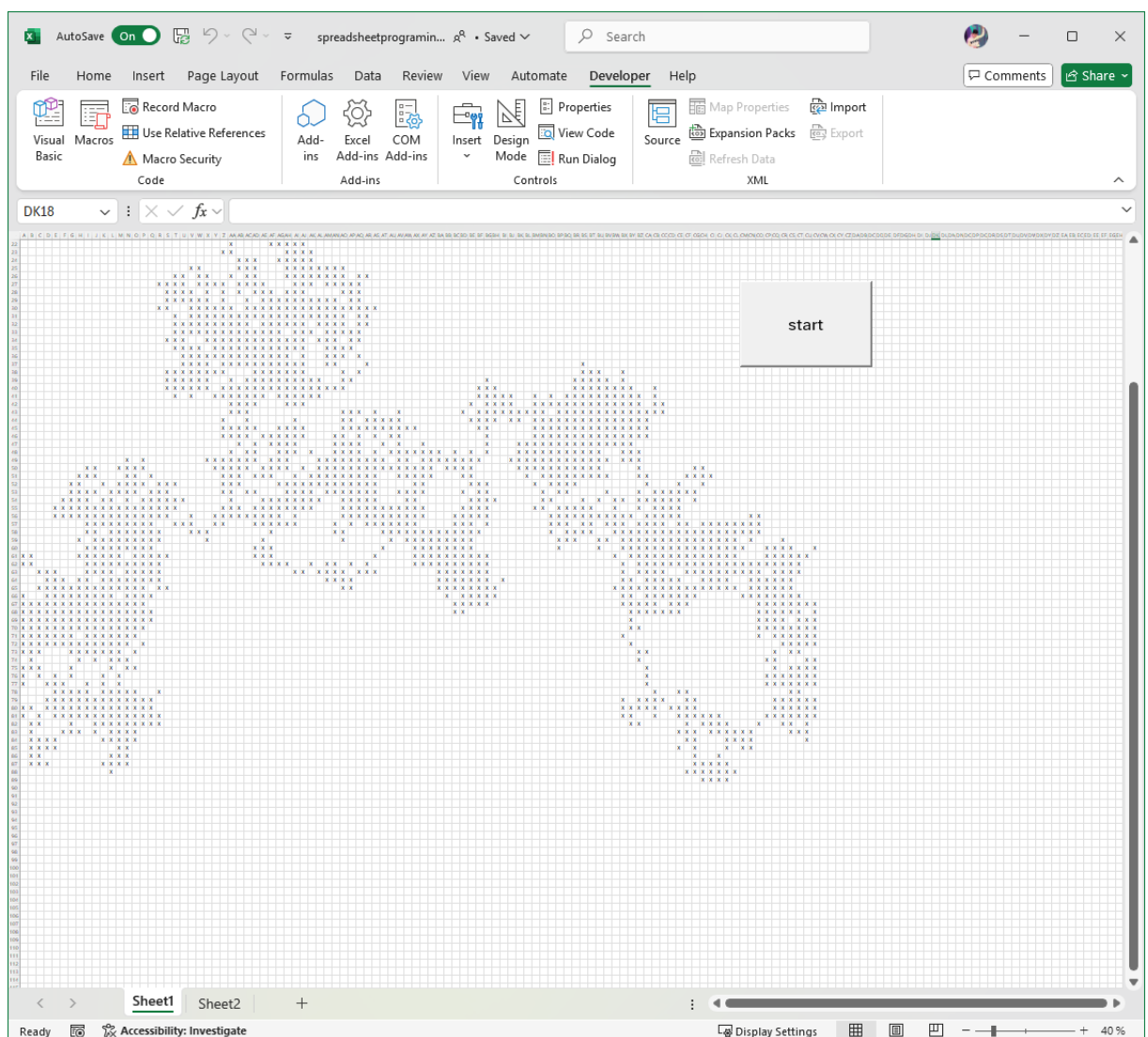
9. Revisits (if enabled):

- If revisits are allowed, the walker may step on the same cell multiple times, and each time they do, an "X" will mark that cell again.

10. Path Completion:

- Once the walker has taken the specified number of steps, the simulation will stop, and the final path will be displayed in the Excel grid.

11. Final Display:



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13. After the walk finishes, you can press the **Start** button in Excel to reset the grid and start a new simulation with different settings.

Troubleshooting & Challenges

1. **Transition to UserForms:** The initial challenge was moving from ThisWorkbook code to using UserForms for better user interaction. Ensuring that all parameters were passed correctly between the UserForm and the main simulation was a learning curve.
2. **Handling Input Fields:** Understanding how to use checkboxes, input fields, and command buttons within the UserForm took some time. Debugging involved correctly linking user inputs with the appropriate simulation logic.
3. **Wrap-Around Functionality:** Implementing the wrap-around feature was tricky, as it required handling boundary conditions and correctly adjusting the walker's position when they reached the grid edges.
4. **Logic Errors:** Debugging the step calculation and ensuring the correct application of revisiting rules was a critical part of the development process. The logic had to ensure that the walker did not incorrectly step out of bounds or backtrack unless allowed by the settings.
5. **UserForm Structure:** This was the first experience working with UserForms, and it took time to structure the functions and ensure the input fields and checkboxes were linked to the main program's logic.

Future Enhancements

1. **Multiple Simulations:** Allow running more than one simulation at a time to compare different random walks.
2. **Multiple Starting Points:** Let users choose more than one starting point for the walk.
3. **Better Usability:** Make the form more user-friendly with clearer labels, easy navigation, and error messages when needed.

Resources for Learning

- [Bilyal Khassenov's YouTube Channel](#): When I started creating this project, I had no prior knowledge of spreadsheet programming. After searching on YouTube, I found the channel Bilyal Khassenov, which provided me with valuable tutorials and insights on VBA programming. Although there aren't specific tutorials for the random walk program, the channel helped me develop a foundational understanding of spreadsheet programming.
- [Basic Random Walk Concepts \(1D & 2D\)](#): I also learned the basics of 1D and 2D random walks from this video, which helped me understand the

underlying concepts of movement within a grid and how they could be applied to my project.

- **Professor Rafal Zdunczyk:** Throughout the development of the project, my professor, Rafal Zdunczyk, played an instrumental role by providing real tasks and guiding me on how to approach the project effectively. His assistance was crucial for putting the project together in a structured way.